

Vincent Mallet

TENURE-TRACK RESEARCHER · ECOLE DES MINES

60, Boulevard Saint-Michel, Paris, France, 75006

✉ vincent.mallet@minesparis.psl.eu | 🏠 vincentx15.github.io | 📷 [Vincentx15](#)

Curriculum

- 2026 - **Faculty researcher**, CBIO - Mines Paris
- 2024 - 2025 **Tenure-track researcher**, CBIO - Mines Paris
- 2023 - 2024 **Postdoctoral researcher Supervised by Maks Ovsjanikov**, Ecole Polytechnique
- 2022 **Visiting Student with Emanuele Rodolà**, La Sapienza
- 2019 - 2022 **PhD supervised by Michael Nilges and Jean-Philippe Vert**, Institut Pasteur / Mines Paris
- 2018 - 2019 **MSc in Computer Science with Jérôme Waldispühl**, McGill Computational Biology Group
- 2013 - 2019 **MSc in Bioinformatics**, Ecole Polytechnique

Publications

JOURNALS

Structure-based rna drug discovery with deep-learning assisted virtual screening. Juan Guillermo Carvajal Patiño*, Vincent Mallet*, David Becerra, Luis Fernando Niño, Carlos Oliver, and Jérôme Waldispühl. *Nature Communications*, 2025

Indeepnet: a web platform for predicting functional binding sites in proteins using indeep. Fabien Mareuil, Rachel Torchet, Luis Checa Ruano, Vincent Mallet, Michael Nilges, Guillaume Bouvier, and Olivier Sperandio. *Nucleic Acids Research*, page gkaf403, 2025

Finding antibodies in cryo-em maps with crai. Vincent Mallet, Chiara Rapisarda, Hervé Minoux, and Maks Ovsjanikov. *Bioinformatics*, 41(5):btaf157, 2025

Alignscape, displaying sequence similarity using self-organizing maps. Isaac Filella-Merce, Vincent Mallet, Eric Durand, Michael Nilges, Guillaume Bouvier, and Riccardo Pellarin. *Frontiers in Bioinformatics*, 4, 2024

InDeep: 3D fully convolutional neural networks to assist in silico drug design on protein-protein interactions. Vincent Mallet, Luis Checa Ruano, Alexandra Moine Fanel, Michael Nilges, Karen Druart, Guillaume Bouvier, and Olivier Sperandio. *Bioinformatics*, 2022

RNAglib: A python package for RNA 2.5D graphs. Vincent Mallet, Carlos Oliver, Jonathan Broadbent, William L Hamilton, and Jérôme Waldispühl. *Bioinformatics*, 2022

VeRNAL: A Tool for Mining Fuzzy Network Motifs in RNA. Carlos Oliver*, Vincent Mallet*, Pericles Philippopoulos, William L Hamilton, and Jérôme Waldispühl. *Bioinformatics*, 2022

quicksom: Self-Organizing Maps on GPUs for clustering of molecular dynamics trajectories. Vincent Mallet, Michael Nilges, and Guillaume Bouvier. *Bioinformatics*, 2021

Optimol: Optimization of binding affinities in chemical space for drug discovery. Jacques Boitreau*, Vincent Mallet*, Carlos Oliver, and Jérôme Waldispühl. *Journal of Chemical Information and Modeling*, 2020

Augmented base pairing networks encode RNA-small molecule binding preferences. Carlos Oliver, Vincent Mallet, Roman Sarrazin Gendron, Vladimir Reinharz, William L Hamilton, Nicolas Moitessier, and Jérôme Waldispühl. *Nucleic Acids Research*, 2020

CONFERENCES PROCEEDINGS

Atomsurf: Surface representation for learning on protein structures. Vincent Mallet*, Souhaib Attaiki*, Yangyang Miao*, Bruno Correia, and Maks Ovsjanikov. *ICLR*, 2025

Reverse-complement equivariant networks for dna sequences. Vincent Mallet and Jean-Philippe Vert. *Advances in Neural Information Processing Systems*, 2021

WORKSHOPS AND PREPRINTS

Nicolas Portal, Wissam Karroucha, Vincent Mallet, and Massimiliano Bonomi. Learning dynamic protein representations at scale with distograms. In *ICLR Workshop on Learning Meaningful Representations of Life (LMRL)*, 2026

Victor Gertner, Frederic Cazals, and Vincent Mallet. Alphasurf: On-the-fly surface computations for protein representation learning. In *ICLR Workshop on Learning Meaningful Representations of Life (LMRL)*, 2026

Machine learning for rna-targeting drug design. Wissam Karroucha, Carlos Oliver, Veronique Stoven, and Vincent Mallet. *arXiv preprint arXiv:2512.15645*, 2025

Luis Wyss, Vincent Mallet, Wissam Karroucha, Karsten Borgwardt, and Carlos Oliver. A comprehensive library for rna structure-function modeling. In *ICLR Workshop on AI for Nucleic Acids*, 2025

Awards, Fellowships, & Grants

- 2025 **Prairie-PSAI Junior Chair**, 150k€
- 2025 **PSL PhD Fellowships for Victor Gertner**,
- 2024 **3 PhD Fellowships for Wissam Karroucha**, FRM, PSL, ITN Mines
- 2022 **Elise Mobility Award**, ELLIS
- 2019 **2 PhD Fellowship**, INCEPTION Program, AMX

Presentations

- 2026. *RNA Drug Design*. Invited Talk Sanofi Paris.
- 2026. *Equivariant Neural Networks*. Invited Seminar Slavica Jonic, Sorbonne Paris.
- 2025. *Rationalizing machine learning on biomolecules structures*. Keynote LEGO meeting, Nantes.
- 2025. *Machine learning on the structure of biomolecules*. GT Masim, Paris.
- 2025. *Machine learning on the structure of biomolecules*. CBS Montpellier.
- 2024. *Learning on protein structures*. Bits in Bio Paris.
- 2024. *Learning on protein structures, with applications in Cryo-EM*. Keynote Pasteur Symposium on AI.
- 2023. *Modeling structural biology with geometric deep learning*. Data Shapes Seminar, INRIA Paris.
- 2023. *Geometric learning for drug design*. AstraZeneca drug discovery Seminar (online).
- 2023. *RNA $\text{GL}ib$: a platform for learning on RNA structure*. MILA drug discovery Seminar.
- 2023. *Reverse-Complement Equivariant Networks for DNA Sequences*. AI in Structural Biology Workshop.
- 2022. *Representation learning on the structure of biomolecules* GT MASIM, Paris
- 2022. *InDeep: learning for drug design on protein protein interactions*. Workshop on AI in magnetic resonance.
- 2022. *Artificial Intelligence in prediction, determination and analysis of structures*. EMBO workshop: Machine learning for structural biology.

Teaching Experience

- 2025, 2026 **Lecturer**, Institut Pasteur Introduction to deep neural networks
- 2025-2026 **Lecturer**, Dassault Systèmes Foundations of deep learning
- 2024-2026 **Lecturer**, Ecole des Mines Modern drug design with learning approaches
- 2022-2024 **Teacher**, CRI Paris 45 hours of classes and practical sessions on Foundations of AI
- 2022 **Lecturer**, McGill CS Master Geometric learning for drug design
- 2020-2022 **Lecturer**, Université de Paris, Master ISDD Generative models for drug design

Mentoring

SUPERVISION

- 2026- **Alexandre Ducorroy**, M2 intern
- 2025- **Victor Gertner**, PhD Student
- 2025- **Nicolas Portal**, Post-Doc
- 2025- **Ikram Mahmoudi**, Post-Doc
- 2024- **Wissam Karroucha**, PhD student
- 2024- **Victor Laigle**, PhD student

JOINT MENTORSHIP

- 2025- **Baoquan Chen**, PhD student, *with Daniel Probst, WUR*
- 2025- **Loïc Chadoutaud**, PhD student, *with Thomas Walter and Emmanuel Barillot, Institut Curie*
- 2024- **Tian Zhu**, PhD student, *with Bruno Correia, EPFL*
- 2024- **Mrunali Manjrekar**, PhD student, *with Maz Abulnaga, MIT*
- 2024- **Yangyang Miao**, PhD student, *with Bruno Correia, EPFL*
- 2024- **Tamara Künzle**, PhD student, *with Maks Ovsjanikov, Ecole Polytechnique*
- 2022 **Jacques Boitreaud**, Msc student, *with Jérôme Waldispühl, McGill*

Additional responsibilities

Reviewer: Bioinformatics, NAR, Journal of Chemoinformatics, NeurIPS, ICML, Recomb, ECCB...

Associate Editor: Frontiers in Bioinformatics

Organizer: GT Masim (French Bioinformatics Community), AlgoSB Winter school

Member: Ellis network, PRAIRIE Junior Chair

Funding Committee member: ANR TSIA Biologie et Santé 2025 Call

Doctoral Committee member: Arthur Carre, François Thénier, Enzo Marsot, Estelle Bergiron, Francesca Venturi, Mehya Mlaweh

Packages: rnaglib, quicksom, CrAI, RNAmigos2